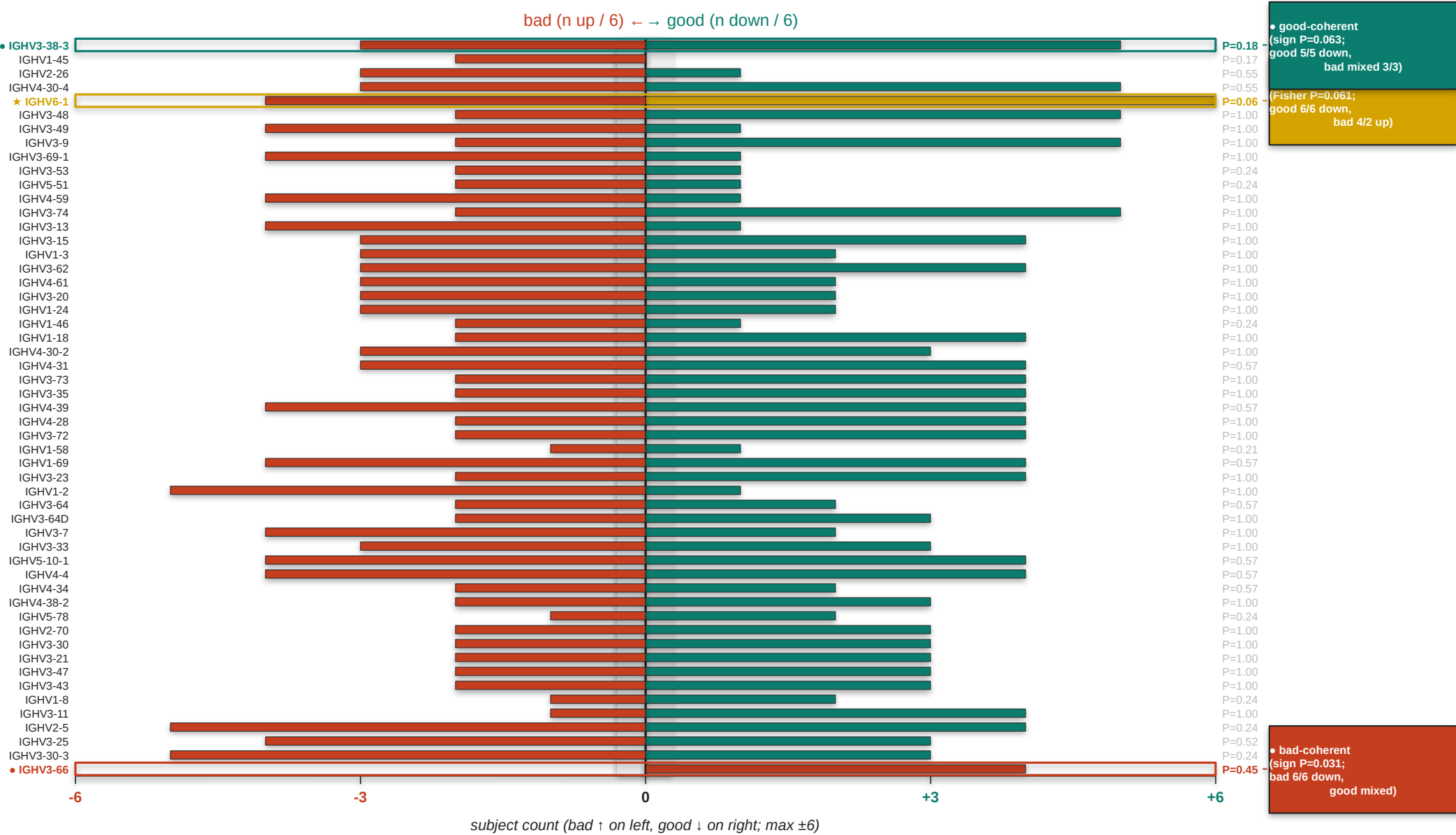


A

IGH V-gene directional-coherence forest — tri-colour highlight by group-level coherence (GOLD=polar-opposite, TEAL=good-only, CORAL=bad-only)

Highlight category

- polar-opposite
- good-coherent only
- bad-coherent only



Sorted by coherence_gap descending. Gray band = $|majority - 0.5| < 3.3/6$ (indistinguishable from chance). Tri-colour highlights mark the three V-genes with individual-level statistical support: IGHV6-1 (between-group Fisher $P=0.061$ + within-good sign $P=0.031$, polar-opposite), IGHV3-38-3 (within-good sign $P=0.063$, good-coherent only), IGHV3-66 (within-good sign $P=0.031$, bad-coherent).

B

Pattern-class scatter (good vs bad majority fraction) — tri-colour highlight on three statistically supported V-genes

